

# Ranjal Arjun Shenoy



☎ +91-9844450000    ✉ ranjalarjunshenoy@gmail.com

🌐 Arjun13-git    🔗 LinkedIn Profile    🌐 Portfolio Website

## Education

### Bachelor of Engineering in Computer Science and Engineering

Sahyadri College of Engineering and Management

2023 – 2027

CGPA: 8.67

## Work Experience

### Software Engineering Intern

March 2026 – September 2026

Datavex.ai Private Limited

- Contributed to developing VexCAD, an AI-powered CAD platform that transforms natural language prompts, sketches, and engineering blueprints into editable 3D models through intelligent design automation.
- Collaborated on developing production-ready modules for an enterprise ERP platform supporting CRM, inventory, billing, accounting, procurement, and business analytics modules.
- Built full-stack features, integrated machine learning capabilities and REST APIs, resolved software defects, and improved application reliability using FastAPI, Next.js, React, TypeScript, PostgreSQL, SQLAlchemy, Docker, and Git.

## Projects

### QML Galaxy Classification & Exoplanet Detection

- Built a flagship Quantum Machine Learning and Deep Learning pipeline for galaxy classification and exoplanet detection, combining attention-based neural networks, variational quantum circuits, image preprocessing, temporal data workflows, and robustness testing on astronomical datasets.
- Technologies:** Python, TensorFlow, Keras, PennyLane, Qiskit, OpenCV, NumPy, Pandas, scikit-learn.

### Sentinel Agents

- Developed a multi-agent AI security platform for static analysis, automated vulnerability detection, secure sandbox execution, exploit validation, and AI-assisted remediation workflows for improving secure code review.
- Technologies:** Python, FastAPI, Tree-sitter, Docker, OpenAI API, Google Gemini API, Next.js, React, Tailwind CSS.

### Project Aether

- Built a Computer Vision based geospatial intelligence platform using YOLOv8 to detect and classify objects in satellite imagery, analyze visual intelligence outputs, and present results through an interactive Streamlit interface.
- Technologies:** Python, Streamlit, YOLOv8, OpenCV, NumPy, Computer Vision.

## Hackathons & Leadership

- AMD Slingshot 2026** — Built **PromptGuard**, an adaptive AI firewall combining rule-based filtering and LLM analysis for real-time prompt security.
- VexStorm'26** — Built **Sentinel Agents**, a multi-agent cybersecurity platform for vulnerability detection, exploit validation, and AI-assisted remediation.
- SIH Internal Hackathon 2025** — Developed solutions for **Transliteration** and **Railway Crossing** problem statements.
- Versathon** — Developed an **Automated Digital KYC and Onboarding Platform** for secure identity verification and onboarding.
- Google DeepMind Hackathon** — Built **Project Aether**, a computer vision platform for satellite imagery analysis.
- SSTH 2024** — **Mentor, Synergia** — Mentored two teams, guiding project development, refinement, and presentation.

## Skills

**Languages:** Java (DSA), Python, HTML

**Frameworks & Libraries:** React.js, Next.js, Node.js, FastAPI, Streamlit, TensorFlow/Keras, Pytorch, OpenCV, NumPy, Pandas, scikit-learn, Selenium, LangChain, LangGraph, PennyLane, Qiskit

**AI / ML:** Deep Learning, Quantum Machine Learning, Computer Vision, Agentic AI, LSTM, YOLOv8, CUDA Toolkit, LLMs

**Databases & Vector Stores:** PostgreSQL, MongoDB, MySQL, ChromaDB, Pinecone

**Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook, Anaconda, Docker, Linux (Endeavour OS)

**Core CS:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Software Engineering, Computer Networks